

1. Projectile motion is most accurately described as the motion of an object that:

- a. Proceeds along a linear trajectory with constant velocity
- b. Maintains uniform motion without acceleration
- c. Follows a circular path due to continuous centripetal force
- d. Moves solely under the influence of a constant gravitational field after launch

2. In classical mechanics, a projectile is defined as an object that:

- a. Restricts its motion entirely to the horizontal direction
- b. Experiences only resistive forces during motion
- c. Is imparted an initial velocity and subsequently moves under the influence of gravity alone
- d. Remains stationary until acted upon by an external force

3. In ideal projectile motion neglecting air resistance, how does the constant gravitational force influence the kinematic properties of the projectile?

- a. By continuously altering the horizontal component of velocity
- b. By producing a uniform acceleration that modifies only the vertical component of velocity
- c. By progressively bringing both velocity components to zero
- d. By changing the intrinsic mass of the projectile during flight

4. What is the most scientifically valid explanation for discrepancies observed between experimentally measured values and theoretically predicted results in a projectile motion experiment?

- a. The realization of perfectly controlled experimental conditions
- b. The inherent inaccuracy of established physical laws
- c. The constancy of gravitational acceleration in all environments
- d. The presence of systematic and random errors, including measurement uncertainties and unmodeled forces such as air resistance

5. When the launch angle is adjusted to 0° relative to the horizontal, the initial velocity vector of the projectile is directed:

- a. Entirely along the positive vertical axis
- b. Exclusively along the horizontal axis
- c. Equally along both horizontal and vertical axes
- d. Along a downward-sloping trajectory

6. Within the coordinate system defined for the projectile motion experiment, the horizontal origin is most appropriately assigned as:

- a. $x = 1$, representing the first measured trial
- b. $x = v$, representing initial velocity
- c. $x = h$, indicating the vertical displacement
- d. $x = 0$, corresponding to the point of projectile release

7. In the determination of the projectile's range, the horizontal displacement is defined as the distance between:

- a. The uppermost point of the launching apparatus and the ground
- b. The final resting point of the projectile only
- c. The horizontal position of release and the point of first ground contact
- d. The apex of the projectile's trajectory and the landing point

8. What is the principal statistical advantage of performing multiple trials and averaging the results in a projectile motion experiment?

- a. It deliberately amplifies random uncertainties in the data
- b. It reduces the overall precision of the measurements
- c. It alters the intrinsic velocity of the projectile
- d. It enhances the reliability and reduces the influence of random errors

9. In the analysis of projectile motion for an object launched horizontally from a known height, the relation

$$y = gx^2 / 2v_0^2$$

is primarily used to determine which of the following quantities when all other variables are measured?

- a. The intrinsic mass of the projectile
- b. The total time of flight directly
- c. The magnitude of the initial horizontal velocity
- d. The launch angle relative to the horizontal

10. Under ideal conditions neglecting air resistance, which expression correctly gives the theoretical horizontal range of a projectile launched with initial speed v_0 at an angle θ ?

- a. $R = gh$, relating gravitational potential energy to height
- b. $R = v_0 t$, valid only without considering vertical motion
- c. $R = v_0^2 \sin 2\theta / g$, derived from kinematic equations
- d. $R = g/v_0$, which lacks dimensional consistency

Answer Key:

1. d
2. c
3. b
4. d
5. b
6. d
7. c
8. d
9. c
10. c